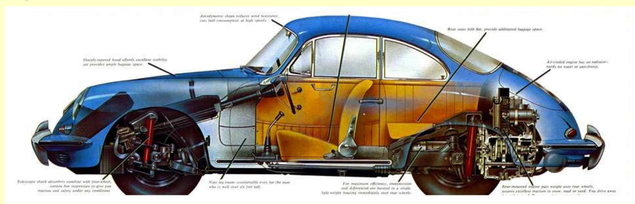
Fluid Mechanics
Project based Learning

Topic : CAR CROSS-SECTION

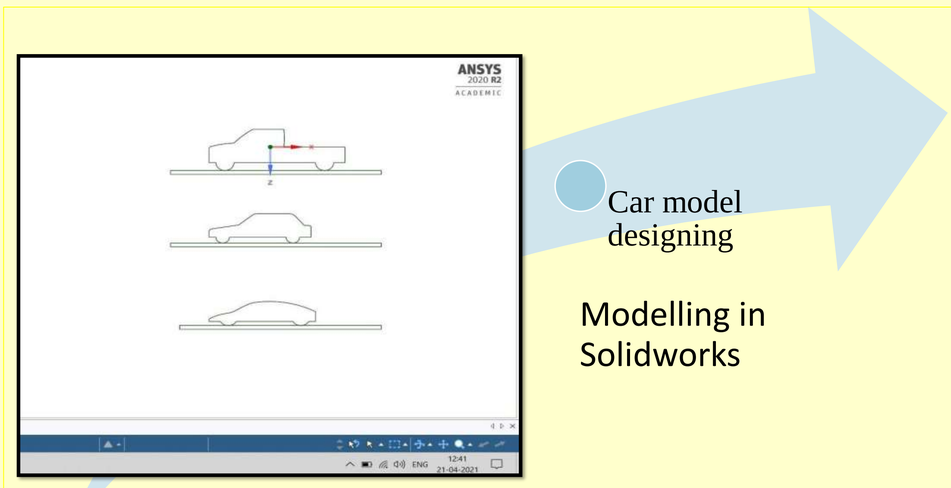
Introduction

Automobiles on the roads today have various shapes and sizes, like sedans, hatchbacks, pick-up trucks, SUVs, sports cars and more. Each car model is optimized for a specific operation; for example, a sports car is designed to optimize the lift and drag forces acting on the car so that it can reach top speeds over a short duration and enhance maneuvering capabilities at high speeds.

Aerodynamic Forces
Thrust - which assists with pushing an object ahead.
Drag - which keeps it down.
Lift - which keeps it airborne.



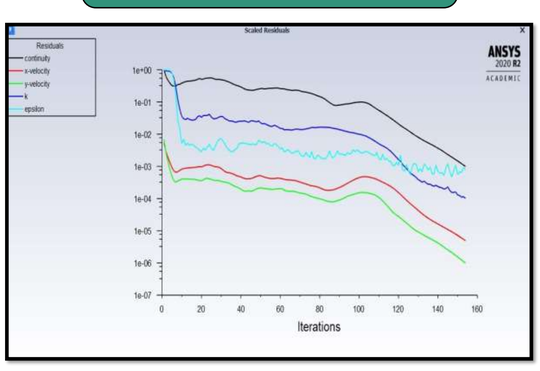
Results

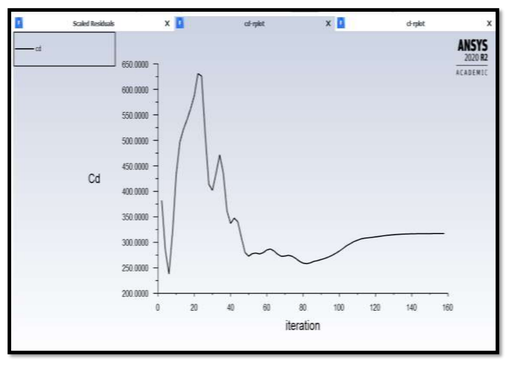


Objective

In this simulation, you will learn how to set up a simulation to analyze the flow over a 2D sedan cross section and to calculate various forces acting on the car. By using "ANSYS FLUENT."

Results





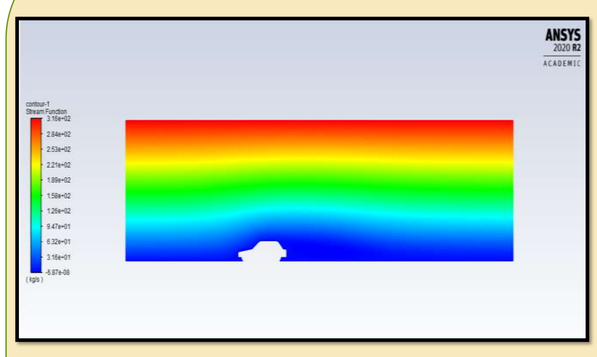
Graph of Drag

Physics set up

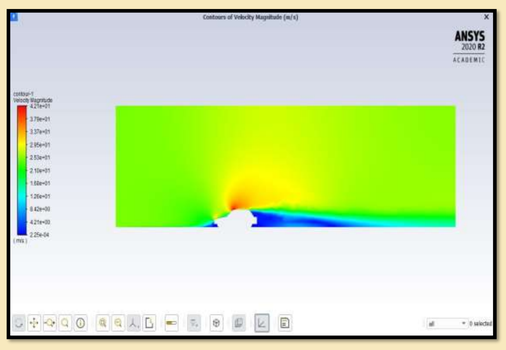
Steps involved in Physics Setup:

1. Use steady state pressure based solver and select all the options.
2.Set viscous model. Use realizable k-epsilon model with enhanced wall treatment and click ok.
3.Set material properties. Use constant density.
4.Define boundary conditions. Set inlet as Velocity- Inlet with Velocity Magnitude of 25 m/s.
Set outlet as Pressure-Outlet and symmetry as Symmetry.

Results



Contours of stream function

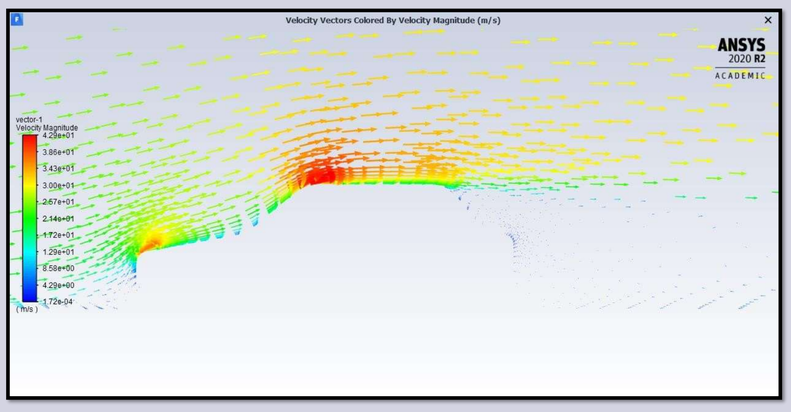


Contours of velocity magnitude

Numerical solution

Set solution methods and controls. Use defaults.
2.Set residuals criteria to 1e-3.
3.Define monitors for drag and lift coefficients.
4.Initialize and fill no. of iterations 500 and run the simulation

Results



Velocity vectors

Team Members :

1. Siddharth Yadav 2. Fakidul Azam
3. Jogesh Bhardwaj 4. Shahbaz Khan
5. Shahzeb Anwar

Instructor : MR. Faisal